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Routing Optimisation for Freight Operations

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Declaration

The work presented in this report is my own work and all references are duly acknowledged.

This work has not been submitted, in whole or in part, in respect of any academic award at Curtin University or elsewhere.

Cem Alpaslan

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Acknowledgments

I would like to say thank you and present my respect to my project supervisors Elham Mardaneh and Andy Zhang and the course coordinator Professor Yong Wu for their help, guidance and support throughout the entire semester. I can not complete the project successfully without their support and guidance.

Vehicle Routing Problems is a topic I am really interested in and have had the opportunity to complete various projects in the past, but I believe this project has improved my knowledge and understanding of how to approach a highly complex vehicle routing problem and provided me with practical experience. I express my gratitude to my supervisors, and to Curtin University for providing this opportunity as a final project unit.

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Abstract

Transportation has been a significant operation for many companies in various industries over the years. Efficient transportation operations have enhanced companies' overall success, customer satisfaction, productivity and total profit. Freight operations such as road, air and maritime play a vital role for effective transportation. Optimizing freight operations has been a crucial research point that has attracted the interest of academicians, mathematicians, engineers and industry professionals. Vehicle routing problems are an essential topic to optimize freight operations in various fields. This project is focused on providing a new solution methodology for a highly specific vehicle routing problem which covers its various extensions. A novel mathematical model is developed and proposed to the audience with practical examples. Lastly, detailed analysis of the results is presented in the report to provide crucial insights into the professionals to optimize their routing strategies.



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1. Introduction

1.1 Problem Description and Background

Nowadays, logistics is an essential function to provide efficient operations for various industries such as transportation, healthcare, mining, manufacturing and agriculture. Logistics is the whole process of planning, monitoring and transferring goods from initial locations to end users (Jenkins 2025). Jenkins (2025) indicates that the global market size of logistics operations will be raised by 4.7% by 2027, reaching a market volume of \$6.6 trillion. Logistics is an inseparable function for almost every company in various industries. Furthermore, companies seek new approaches to improve the efficiency of their logistics operations. Chaitanya (2025) highlights that one of the main benefits of having a robust logistics operation is cost improvement. In addition, well-functioning logistics operation enhances the supply chain monitoring and overall performance of the company (Chaitanya 2025). Moreover, optimized logistics operation minimizes carbon dioxide emissions, air pollution and fuel consumption which maintains environmental sustainability and minimizes environmental damage (Chaitanya 2025). Chaitanya (2025) indicates that avoiding late deliveries improves customer satisfaction and increases the retention rate of the customers. Effects of optimized logistics operations are undeniable. In addition, freight operations are the backbone of logistics operations. McCue (2025) states that freight operations are the main function of supply chains which focus on transporting units to various desired locations. The global supply chain networks of the world encourage companies to expand their freight fleets and adapt their operations into road, air and maritime transportation solutions (McCue 2025). Vehicle Routing Problems (VRP) have been a mandatory research field for many years for optimizing transportation operations. VRP are introduced by mathematicians called George Dantzig and John Ramser in 1959 as the extension of Traveling Salesman Problem (TSP) (Patel 2026). After the introduction of the VRP, academicians, mathematicians and professionals from various industries have focused on providing solution methodologies for its extension. Furthermore, the main difference between VRP and TSP is VRP focuses on providing optimal routing combinations for more than one vehicle with respect to real life constraints while TSP focuses on finding the optimal routing sequence for one vehicle (Patel 2026). Patel (2026) points out that the main disadvantage of VRP is its complexity. VRP is a well-known NP-hard problem; therefore, adding real life constraints such as time windows, capacities and multiple depots and growing the dataset enhances its complexity (Patel 2026). Mathematical programming has been the primary solution technique for vehicle routing problems for many years (Leite 2023). Since the

performances of the commercial solver tools and their underlying algorithms have improved significantly over the years, mathematical modelling has been used as the solution approach for small instances for many years (Patel 2025). Furthermore, algorithm development, decomposition techniques, machine-learning methods, heuristics and metaheuristics have been used as the primary solution methodologies for real life large instance scenarios and they have been providing feasible solutions within computationally efficient time frames (Patel 2025).

The main focus in this research is to analyze and provide a solution methodology for a highly specific vehicle routing problem. This project is inspired by a real-life freight optimisation problem that has reached the Centre for Optimisation and Decision Science at Curtin University. The project couldn't be conducted due to certain complications but my supervisor recommended using this project as a template to my final project unit. The project focuses on optimizing short-term freight planning strategies. In addition, project is designed to serve various customer locations with a single depot that can be used as a temporary storing point. Planning and shipment horizon of the problem is limited to two working days. Customer locations consist of pick-up and delivery locations and pick-ups must be completed before the deliveries. Moreover, there are four different shipment scenarios in the shipment horizon. First one is that pickups and deliveries can be made on the same day. For instance, the pickup of the shipment can be done on the first day of the horizon and delivery can be also done on the first day of the horizon. Similarly, pickup can be completed on the second day and the delivery of the shipment can be executed on the second day of the horizon as well. Secondly, pickups and deliveries can be completed on different days. For example, pickup of the shipment can be done on the first day of the horizon but delivery can be executed on the second day. Third scenario covers executing only deliveries of the shipments. Pickups of some of the shipments have already been completed before the horizon and stay in the depot, this scenario covers executing the deliveries of those shipments on day one or day two. Last scenario is performing only the pickups of the shipments and not completing the deliveries of the shipments within two days and transferring them to further horizons. Main objective of the project is to develop a mathematical model to minimize the total cost which consist of fixed vehicle renting cost, distance per kilometer cost (fuel cost and driver wage) and cross docking (load exchange) at the depot cost. Furthermore, this project is examined to find how to route the vehicles, which vehicles to use, how to allocate shipments to the vehicles and sequencing of the shipment locations. This project also involves various additional considerations. It will be assumed that vehicles will be outsourced and it should be identified what vehicles should be hired.

Furthermore, the vehicle options provided consist of a heterogeneous fleet and there are three types of vehicles which are small, medium and large and all of them have different capacities and costs. Moreover, as mentioned every vehicle has different capacity options and the total cargo loaded onto the vehicles cannot exceed the vehicle capacity. Furthermore, depot has different working hours in the two-day shipment horizon and every node location has designated time windows. Vehicles should visit the nodes within those time window ranges and perform the necessary operations without any time window violation. After completing pickups and deliveries vehicle load should be updated and monitored accurately. In addition, each vehicle has a daily distance limit and must turn back to the depot without violating the limitation. Key points of this problem that should be concentrated on are multi-trip options, cross-docking and inventory management. A vehicle can perform more than one trip and may visit the depot multiple times during the day. Furthermore, cross docking (load exchange) option is another crucial element to be performed. A shipment pair can be performed by different vehicles. For instance, one vehicle can perform the pickup of the shipment and another vehicle can execute the delivery of the shipment. A vehicle can visit the depot and unload its cargo and another vehicle can load that amount and continue its trip. Lastly, at the end of the first day total cargo unloaded to the depot must be used on the second day of the planning horizon; therefore, the inventory should be transferred and managed accurately to perform uninterrupted routing operations in the two-day shipment horizon.

1.2 Objectives, Significances and Contributions of the Project

The primary objective of this project is to analyze previous vehicle routing problems and their solution approaches to provide a new solution methodology for a highly specific vehicle routing problem. Main target of this project is to investigate the problem requirements and develop a mathematical formulation to find the optimal routing strategies with minimizing the total cost. Furthermore, using advanced Gurobi settings and parameters and gap reduction techniques such as arc-based tight Big-M selection and removing unnecessary arcs will provide practical insights to the developers. Additionally, planned future work will cover designing a metaheuristic algorithm specifically for this mathematical formulation will enhance the contribution of the project. This project seeks to:

- Analyze and do research about the previous similar vehicle routing problems, its extensions and their solution methodologies.

- Establish a mathematical model to provide a new solution methodology for this vehicle routing problem.
- Develop a mathematical model code of the established mathematical formulation.
- Provide theoretical and practical outcomes for the freight companies to optimize their routing strategies.

This project is significant because it will provide both academic and practical contributions. The developed mathematical model will be available for the interested audience and it can be used for further developments or as a template for other vehicle routing problem extensions. Additionally, it can easily be converted into air and maritime freight operation problems by the experts. Lastly, developed mathematical model code and the metaheuristic algorithm code to be developed will be available on my GitHub profile; therefore, it can be used by developers for further practical contributions.

1.3 Literature Review (Review and Limitations of Previous Studies)

Vehicle routing problems have always been a crucial topic in operations research. Engineers, mathematicians, operations research scientists and academicians have focused on providing solution methodologies and algorithms for various vehicle routing problems and their extensions. Various mathematical techniques such as integer programming, linear programming, mixed integer linear programming, heuristic and metaheuristic algorithms, decomposition methods and column generation have been used for many years to provide optimisation strategies to road, air and maritime freight operations. To develop my mathematical model, I have investigated various articles and different types of vehicle routing problems, because this project covers several VRP extensions and constraints together. Santos et al. (2012, 1085) focused on providing a mathematical formulation to pick up and delivery vehicle routing problem with cross docking. Developed mathematical formulation is a linear programming model and objective is to minimize the cross-docking costs (Santos et al. 2012, 1087). I have used this article to design my cross-docking scenarios and constraints. However main point lacking in the article was that it did not cover multiple trip options. Furthermore, Keskin and Çatay (2016, 111) concentrated on providing a solution methodology for electric vehicle routing problems. The research focused on formulating a mathematical model to minimize the total distance traveled for electric vehicles and re-charge stations and strengthened with a neighborhood search algorithm (Keskin and Çatay 2016, 114). Although the VRP concept in this article was completely different, I have used the time window constraints to

create my time window restrictions of my problem. Moreover, Maknoon and Laporte (2016, 254) addressed a vehicle routing problem with pickups and deliveries and cross docking options in their study. The developed mathematical model in the article designed to find the optimal routing strategy to minimize the total cost (Maknoon and Laporte 2016, 256). Even though this research has helped my study, there was a crucial limitation because it was designed for visiting multiple cross docking points. In my project there is a single depot and serves as a cross-docking point. Lastly, Niyomphon and Wissittipanich (2022, 1) focused on an urban delivery transportation problem to provide routing strategies for minimizing the total cost of the operation. This study was a beneficial source for my mathematical formulation because it has demonstrated that creating a separate set in the mathematical model is necessary to satisfy multiple trip options.

2. Methodology

2.1 Mathematical Modelling

To solve our highly complex and specific vehicle routing problem, mathematical modelling was used as the solution methodology in this project. A mixed-integer linear programming mathematical formulation was developed by defining the sets, parameters, decision variables, objective function and constraints.

Assumptions:

- It is assumed that there will be no split deliveries in this problem and it is not allowed in the mathematical formulation.
- It is also assumed that all vehicles must turn back to depot after completing their operations in each day. It is not an open vehicle routing problem.
- It is assumed that at the end of each trip and each day, vehicle turns back to the depot and may partially load and unload its cargo. Cross-docking is allowed in this problem.

Sets:

$T = \{1,2\}$: Set of days

$K = \{1,2,3\}$: Set of vehicles, 1 is small vehicle, 2 is medium vehicle and 3 is large vehicle

$R = 1,2, \dots, Rmax$: Set of trips

$N = \{0\} \cup P \cup D$: Set of nodes where 0 denotes the depot, P pickup nodes, D delivery nodes

$A = \{(i, j) \in N \times N : i \neq j\}$: Set of arcs

S = Set of shipments where $s \in S$, $p(s) \in P$ is the pickup node and $d(s) \in D$ is the delivery node

Parameters:

D_{ij} : Distance from node i to node j

T_{ij} : Travel time from node i to node j

S_i : Service duration time at node i

E_i : Earliest service start time at node i

L_i : Latest service start time at node i

E_{0t} : Earliest opening time of the depot on day t

L_{0t} : Latest closing time of the depot on day t

Q_k : Cargo capacity of vehicle k

C^{dep} : Storage capacity of the depot

F_k : Fixed daily rental cost of vehicle k

A_k : Distance per kilometer cost (fuel and driver wage) for vehicle k

H_{kt} : Maximum daily travel distance limit of vehicle k on day t

τ : Reload / handling time at the depot between two trips

q_s : Amount of shipment s

C^{cd} : Cost incurred when a shipment changes vehicles at the depot (cross-docking cost)

I_0 : Initial inventory level at the depot before the shipment horizon

$\alpha_s \in \{0,1\}$: Pickup status parameter

$\alpha_s = 1$: Shipment already picked up before the horizon and ready in the depot

$\alpha_s = 0$: Pickup must be done within the horizon

Δ_i : Net load change at node i

M : Large positive constant (Big-M)

Decision Variables:

$x_{ij}^{ktr} \in \{0,1\}$: 1 if vehicle k travels from node i to node j on trip r on day t ; 0 otherwise

$w^{ktr} \in \{0,1\}$: 1 if trip r of vehicle k is active on day t ; 0 otherwise

$v_{kt} \in \{0,1\}$: 1 if vehicle k is rented on day t ; 0 otherwise

$b_i^{ktr} \geq 0$: Service start time at node i for vehicle k on trip r on day t

$u_i^{ktr} \geq 0$: Load of vehicle k after leaving node i on trip r on day t

$l_{out}^{ktr} \geq 0$: Load taken from the depot at the start of trip r by vehicle k on day t

$l_{in}^{ktr} \geq 0$: Load dropped off to the depot at the end of trip r by vehicle k on day t

$I_t \geq 0$: Inventory level at the depot at the end of day t

$\delta_s \in \{0,1\}$: 1 if shipment s changes vehicles at the depot (cross-docking); 0 otherwise

Mathematical Formulation:

$$\min Z = \sum_{t \in T} \sum_{k \in K} F_k v_{kt} + \sum_{t \in T} \sum_{k \in K} \sum_{r \in R} \sum_{(i,j) \in A} A_k D_{ij} x_{ij}^{ktr} + C^{cd} \left(\sum_{s \in S} \delta_s \right) \quad (1)$$

The objective function minimizes the total cost which includes fixed daily rental cost, distance per kilometer cost and cross-docking (load exchange) cost.

$$w^{ktr} \leq v_{kt}, \quad \forall k \in K, t \in T, r \in R \quad (2)$$

This constraint ensures that a vehicle must be hired if a trip is active. It avoids creating trips with unrented vehicles.

$$w^{ktr} \leq w^{ktr'}, \quad \forall k \in K, t \in T, r, r' \in R: r' \leq r \quad (3)$$

This constraint enforces sequential trip activation. I put this constraint to enhance the gap reduction because it prevents creating solutions with different trip index combinations.

$$\sum_{j \in N \setminus \{0\}} x_{0j}^{ktr} = w^{ktr}, \quad \forall k \in K, t \in T, r \in R \quad (4)$$

This constraint satisfies that a trip must start from the depot.

$$\sum_{i \in N \setminus \{0\}} x_{i0}^{ktr} = w^{ktr}, \quad \forall k \in K, t \in T, r \in R \quad (5)$$

This constraint satisfies that a trip must end at the depot.

$$\sum_{j \in N} x_{ij}^{ktr} - \sum_{j \in N} x_{ji}^{ktr} = 0, \quad \forall i \in N \setminus \{0\}, k \in K, t \in T, r \in R \quad (6)$$

This constraint above is a flow balance constraint. It ensures route continuity by balancing inflows and outflows.

$$\sum_{k \in K} \sum_{t \in T} \sum_{r \in R} \sum_{j \in N} x_{p(s),j}^{ktr} = 1, \quad \forall s \in S: \alpha_s = 0 \quad (7)$$

This constraint ensures that if the pickup of shipment s is not waiting at the depot which means ($\alpha_s = 0$) its pickup node must be visited exactly once within the horizon.

$$\sum_{k \in K} \sum_{t \in T} \sum_{r \in R} \sum_{j \in N} x_{d(s),j}^{ktr} = 1, \quad \forall s \in S \quad (8)$$

This constraint is paired with constraint (7) and it satisfies that delivery node of shipment s must be visited exactly one.

$$0 \leq l_{out}^{ktr} \leq Q_k w^{ktr}, \quad \forall k \in K, t \in T, r \in R \quad (9)$$

This constraint is for satisfying the load taken from the depot at the beginning of a trip cannot exceed the vehicle cargo capacity.

$$0 \leq l_{in}^{ktr} \leq Q_k w^{ktr}, \quad \forall k \in K, t \in T, r \in R \quad (10)$$

This constraint is paired with constraint (9) and it ensures that load dropped off to the depot at the end of a trip cannot exceed the vehicle cargo capacity.

$$u_0^{kt1} = l_{out}^{kt1}, \quad \forall k \in K, t \in T \quad (11)$$

This constraint above is to ensure load satisfaction. For the first trip on each day, load of a vehicle should be equal to the pallets loaded from the depot.

$$u_0^{ktr} = l_{out}^{ktr}, \quad \forall k \in K, t \in T, r \in R: r > 1 \quad (12)$$

This constraint is paired with constraint (11) and provides the same logic for other possible trips (except the first trip on each day because it has been ensured by constraint 11).

$$\begin{aligned} \sum_{k \in K} \sum_{r \in R} u_0^{ktr} = & \sum_{k \in K} \sum_{r \in R} u_0^{k(t-1)r} + \sum_{k \in K} \sum_{r \in R} \sum_{i \in N} \sum_{j \in N \setminus \{i\}} \Delta_j x_{ij}^{k(t-1)r} - \sum_{k \in K} \sum_{r \in R} l_{in}^{k(t-1)r} \\ & + \sum_{k \in K} \sum_{r \in R} l_{out}^{ktr}, \quad \forall t \in T \end{aligned} \quad (13)$$

This constraint above ensures the balance of the daily load that has been departed from the depot across all types of vehicles and trips.

$$u_j^{ktr} \geq u_0^{ktr} + \Delta_j - M(1 - x_{0j}^{ktr}), \quad \forall j \in N \setminus \{0\}, k \in K, t \in T, r \in R \quad (14)$$

I designed this constraint for covering node to depot scenario. It ensures the load update of the vehicle after visiting the first pickup or delivery node after departing from the depot. Moreover, it works as a lower bound.

$$u_j^{ktr} \leq u_0^{ktr} + \Delta_j + M(1 - x_{0j}^{ktr}), \quad \forall j \in N \setminus \{0\}, k \in K, t \in T, r \in R \quad (15)$$

This constraint is paired with constraint (14) and covers node to depot scenario. It works as an upper bound for updating the load of the vehicle after visiting the first pickup or delivery point after departing from the depot.

$$u_j^{ktr} \geq u_i^{ktr} + \Delta_j - M(1 - x_{ij}^{ktr}), \quad \forall (i, j) \in A, j \neq 0, k \in K, t \in T, r \in R \quad (16)$$

This constraint is developed for covering node-to-node scenarios. It considers each pickup and delivery node throughout the tour and updates the load of the vehicle accurately after each visit. It ensures the lower bound.

$$u_j^{ktr} \leq u_i^{ktr} + \Delta_j + M(1 - x_{ij}^{ktr}), \quad \forall (i, j) \in A, j \neq 0, k \in K, t \in T, r \in R \quad (17)$$

This constraint is paired with constraint (16) and designed for node-to-node scenarios as well. It ensures the upper bound of the vehicle after each node visit throughout the tour.

$$\begin{aligned} u_i^{ktr} - M(1 - x_{i0}^{ktr}) & \leq l_{in}^{ktr} \leq u_i^{ktr} + M(1 - x_{i0}^{ktr}), \\ \forall i \in N \setminus \{0\}, k \in K, t \in T, r \in R \end{aligned} \quad (18)$$

The constraint above ensures that load dropped off at the depot must be equal to the load of the vehicle at the last visited node in the trip.

$$0 \leq u_i^{ktr} \leq Q_k w^{ktr}, \quad \forall i \in N, k \in K, t \in T, r \in R \quad (19)$$

This constraint is developed to satisfy the general capacity. It ensures that load of the vehicle cannot exceed the total cargo capacity of the vehicle.

$$u_j^{ktr} \leq Q_k \left(\sum_{i \in N} x_{ij}^{ktr} \right), \quad \forall j \in N \setminus \{0\}, k \in K, t \in T, r \in R \quad (20)$$

I put this constraint to force the load to be zero if a vehicle does not visit a node. It is a logical constraint.

$$b_j^{ktr} \geq b_i^{ktr} + S_i + T_{ij} - M(1 - x_{ij}^{ktr}), \quad \forall (i, j) \in A, j \neq 0, k \in K, t \in T, r \in R \quad (21)$$

This constraint is designed for ensuring the time relationship between each node. It considers service start times, travel times and service duration times.

$$E_i - M \left(1 - \sum_{j \in N} x_{ij}^{ktr} \right) \leq b_i^{ktr} \leq L_i + M \left(1 - \sum_{j \in N} x_{ij}^{ktr} \right), \\ \forall i \in N \setminus \{0\}, k \in K, t \in T, r \in R \quad (22)$$

This constraint ensures the time window requirements. Service at node i must start within its time window values.

$$E_{0t} w^{ktr} \leq b_0^{ktr} \leq L_{0t} w^{ktr}, \quad \forall k \in K, t \in T, r \in R \quad (23)$$

This constraint is designed for depot's time window restrictions. It ensures that every active trip must operate between the depot's operating hours.

$$b_0^{ktr'} \geq b_i^{ktr} + S_i + T_{i0} + \tau - M(2 - x_{i0}^{ktr} - w^{ktr'}), \\ \forall i \in N \setminus \{0\}, k \in K, t \in T, r, r' \in R, r' > r \quad (24)$$

This constraint is developed for sequencing trips accurately. It ensures that trip's depot arrival time must be completed before its departure time.

$$b_0^{ktr} + \sum_{i \in N \setminus \{0\}} S_i \left(\sum_{j \in N} x_{ij}^{ktr} \right) + \sum_{i \in N} \sum_{j \in N} T_{ij} x_{ij}^{ktr} \leq L_{0t}, \quad \forall k \in K, t \in T, r \in R \quad (25)$$

This constraint satisfies that total time of trip r must be less than depot's closing time.

$$\sum_{t \in T} t \left(\sum_{k \in K} \sum_{r \in R} \sum_{j \in N} x_{p(s),j}^{ktr} \right) \leq \sum_{t \in T} t \left(\sum_{k \in K} \sum_{r \in R} \sum_{j \in N} x_{d(s),j}^{ktr} \right), \quad \forall s \in S, \alpha_s = 0 \quad (26)$$

This constraint is a time-based predecessor constraint. It ensures that pickup of a shipment s must be completed before its delivery day.

$$b_{p(s)}^{ktr} \leq b_{d(s)}^{ktr} + M \left(2 - \sum_{j \in N} x_{p(s),j}^{ktr} - \sum_{j \in N} x_{d(s),j}^{ktr} \right), \forall s \in S, \alpha_s = 0, k \in K, t \in T, r \in R \quad (27)$$

This constraint is a time-based predecessor constraint as well. It ensures that if a pickup and delivery of a shipment is performed on same trip and with same vehicle, its pickup up must be made before its delivery.

$$b_0^{k_dtr_d} \geq b_i^{k_ptr_p} + S_i + T_{i0} - M \left(1 - x_{i0}^{k_ptr_p} \right) - M \left(2 - \sum_{j \neq p(s)} x_{p(s),j}^{k_ptr_p} - \sum_{j \neq d(s)} x_{d(s),j}^{k_dtr_d} \right),$$

$$\forall s \in S, \alpha_s = 0, i \in N \setminus \{0\}, t \in T, k_p, k_d \in K, r_p, r_d \in R \quad (28)$$

This constraint is also a time-based constraint. It ensures the temporal ordering of same day cross docking scenarios. If a shipment's pickup and delivery is performed on the same day, the trip's depot arrival time must be completed before its departure time.

$$\sum_{r \in R} \sum_{(i,j) \in A} D_{ij} x_{ij}^{ktr} \leq H_{kt}, \quad \forall k \in K, t \in T \quad (29)$$

This constraint ensures that a vehicle cannot travel more than its restricted total distance limitation.

$$I_0 = \sum_{s \in S} \alpha_s q_s, \quad (30)$$

This constraint calculates the initial inventory level of the depot. It considers the pickups that have already been done before the planning horizon.

$$I_t = I_{t-1} + \sum_{s \in S: \alpha_s = 0} q_s \left(\sum_{k \in K} \sum_{r \in R} \sum_{j \in N} x_{p(s),j}^{ktr} \right) - \sum_{s \in S} q_s \left(\sum_{k \in K} \sum_{r \in R} \sum_{j \in N} x_{d(s),j}^{ktr} \right), \forall t \in T \quad (31)$$

This constraint is designed for ensuring the depot's daily inventory balance.

$$\sum_{k \in K} \sum_{r \in R} l_{out}^{ktr} \leq I_{t-1} + \sum_{s \in S: \alpha_s = 0} q_s \left(\sum_{k \in K} \sum_{r \in R} \sum_{j \in N} x_{p(s),j}^{ktr} \right), \quad \forall t \in T \quad (32)$$

This constraint calculates the current inventory level and it ensures that the load taken from the depot on day t cannot exceed depot's available capacity.

$$I_t \leq C^{dep}, \quad \forall t \in T \quad (33)$$

This constraint is developed for limiting depot's inventory. It ensures that inventory level of the depot at the end of each day t cannot exceed the total storage capacity of depot.

$$\delta_s \geq \left(\sum_{r \in R} \sum_{j \in N} x_{p(s),j}^{ktr} \right) - \left(\sum_{r \in R} \sum_{j \in N} x_{d(s),j}^{ktr} \right), \quad \forall s \in S, \alpha_s = 0, k \in K, t \in T \quad (34)$$

$$\delta_s \geq \left(\sum_{r \in R} \sum_{j \in N} x_{d(s),j}^{ktr} \right) - \left(\sum_{r \in R} \sum_{j \in N} x_{p(s),j}^{ktr} \right), \quad \forall s \in S, \alpha_s = 0, k \in K, t \in T \quad (35)$$

These two constraints above are paired constraints. Constraints (34) and (35) are designed for capturing same day with different vehicles cross docking scenario.

$$\delta_s \geq \left(\sum_{r \in R} \sum_{j \in N} x_{p(s),j}^{ktr} \right) - \left(\sum_{r \in R} \sum_{j \in N} x_{d(s),j}^{kt'r'} \right), \forall s \in S, \alpha_s = 0, k \in K, t \in T, t' \in T, t' > t \quad (36)$$

$$\delta_s \geq \left(\sum_{r \in R} \sum_{j \in N} x_{d(s),j}^{kt'r'} \right) - \left(\sum_{r \in R} \sum_{j \in N} x_{p(s),j}^{ktr} \right), \forall s \in S, \alpha_s = 0, k \in K, t \in T, t' \in T, t' > t \quad (37)$$

These two constraints are paired and developed for cross-docking scenarios as well. They capture different days with same vehicles and different days with different vehicles cross docking options.

$$\delta_s \leq 2 - \left(\sum_{r \in R} \sum_{j \in N} x_{p(s),j}^{ktr} \right) - \left(\sum_{r \in R} \sum_{j \in N} x_{d(s),j}^{ktr} \right), \forall s \in S, \alpha_s = 0, k \in K, t \in T \quad (38)$$

This constraint is designed for non-cross docking scenario. It avoids cross docking if the pickup and delivery of a shipment is performed by the same vehicle.

$$l_{in}^{ktr} \geq q_s \left(\sum_{j \in N} x_{p(s),j}^{ktr} \right) - M(1 - \delta_s), \quad \forall s \in S, k \in K, t \in T, r \in R \quad (39)$$

This constraint is developed for cross docking scenario. If a pickup of a shipment is made and its delivery will be performed with a different vehicle (cross-docking), the load of the vehicle must be dropped off to the depot after the pickup operation so another vehicle can perform the delivery operation.

$$l_{out}^{ktr} \geq q_s \left(\sum_{j \in N} x_{d(s),j}^{ktr} \right) - M(1 - \delta_s), \quad \forall s \in S, k \in K, t \in T, r \in R \quad (40)$$

This constraint is a pair constraint with constraint (39). This constraint satisfies that if cross docking will be performed, assigned vehicle for the delivery operation must have collected the load to be delivered from the depot at the start of the trip.

$$l_{out}^{ktr} \geq \sum_{s \in S: \alpha_s = 1} q_s \left(\sum_{j \in N} x_{d(s),j}^{ktr} \right), \quad \forall k \in K, t \in T, r \in R \quad (41)$$

This constraint above focuses on the shipments that have already been picked up before the horizon. It ensures that vehicle must have loaded the amount from the depot before starting its trip to perform the delivery operations of those shipments.

$$\delta_s = 0, \quad \forall s \in S, \alpha_s = 1 \quad (42)$$

This constraint is a logic constraint. If a pickup is performed before the planning horizon then cross docking is impossible because there will be no pickup needed for those shipments. This constraint ensures that function.

Domain Constraints:

$$x_{ij}^{ktr} \in \{0,1\}, \quad \forall (i,j) \in A, k \in K, t \in T, r \in R \quad (43)$$

$$w^{ktr} \in \{0,1\}, \quad \forall k \in K, t \in T, r \in R \quad (44)$$

$$v_{kt} \in \{0,1\}, \quad \forall k \in K, t \in T \quad (45)$$

$$\delta_s \in \{0,1\}, \quad \forall s \in S \quad (46)$$

$$b_i^{ktr} \geq 0, \quad \forall i \in N, k \in K, t \in T, r \in R \quad (47)$$

$$u_i^{ktr} \geq 0, \quad \forall i \in N, k \in K, t \in T, r \in R \quad (48)$$

$$l_{out}^{ktr} \geq 0, \quad \forall k \in K, t \in T, r \in R \quad (49)$$

$$l_{in}^{ktr} \geq 0, \quad \forall k \in K, t \in T, r \in R \quad (50)$$

$$I_t \geq 0, \quad \forall t \in T \quad (51)$$

Results

3.1 Performance Optimisation Techniques and Data Generation

System Specifications:

Firstly, Python was used as the programming language and Gurobi was used as the commercial optimisation package to implement and solve the developed mathematical model. Visual Studio Code and Spyder were used as the Integrated Development Environments (IDE) for this project. Although Python has released its latest version 3.14, version 3.12 was used in this project. Even though the latest version of the Gurobi solver tool is 13.0.2, version 12.0.1 was selected to solve the developed mathematical model. Furthermore, full academic license of Gurobi was obtained with my student email and employed without any usage limitations. Moreover, the code was run on my personal computer which has 11th Gen Intel® Core™ i5-11400H CPU (2.70 GHz, 6 cores, 12 threads), 16 GB DDR4 RAM (3200 MT/s) and NVIDIA GeForce RTX 3050 Ti Laptop GPU with 4 GB GDDR6 VRAM.

Performance Optimisation Techniques and Input Parameters:

Vehicle Routing Problems are well-known NP-hard problems; therefore, optimality gap is a crucial point that has been focused on by researchers and developers for many years. Being an NP-hard problem is the main factor of gap reduction, but other factors such as problem complexity and the size of the dataset are very crucial as well. Some mathematical techniques have been widely used to accelerate the gap reduction. Furthermore, commercial optimisation solver companies have presented their advanced algorithms and settings to the users. To accelerate the gap reduction, I used mathematical and solver-based techniques in this project. Firstly, rather than selecting a fixed large one single big-M value, I defined separate big-M's for each constraint in the mathematical model that consists big-M. Furthermore, I calculated the mathematical minimum value of the big-M's for each constraint. Main advantage of this approach was to accelerate the gap reduction by strengthening the LP relaxation, increasing the lower bound and tightening the Branch and Bound operations. Moreover, another crucial mathematical technique in this project was arc pre-elimination. Time window values and depot operation hours were evaluated and after the necessary range calculations the infeasible arcs that are out of that range were deleted by setting their upper bounds to zero. This mathematical technique reduced the size of the model by eliminating the unnecessary parameters and

increased the LP relaxation. Lastly, as shown in figure below Gurobi's advanced settings options were used to enhance the gap reduction of the mathematical formulation code.

```
# Adding the optimisation parameters, settings
VRP.setParam("TimeLimit", 7200) # Time limit, decide the time
VRP.setParam("MIPFocus", 2) # 0 balance, 1 focuses on finding a feasible solution (incumbent), 2 focus
VRP.setParam("Heuristics", 0.3) # Effort given to heuristics; 0 closed, 1 max
VRP.setParam("NoRelHeurTime", 180) # First NoRel Heuristic then B&B, decide the time, now it is 180 seconds fo
VRP.setParam("RINS", 10) # Relaxation Induced Neighborhood Search, it tries 10 times, it improves th
VRP.setParam("Presolve", 2) # 0 closed, 1 conservative, 2 aggressive; it tightens bounds
VRP.setParam("Cuts", 3) # Level of cuts: 0 closed, 1 low, 2 medium, 3 aggressive
VRP.setParam("ImproveStartTime", 6300) # It makes MIP focus 3 after 6300 seconds
VRP.setParam("Symmetry", 2) # Detects and removes symmetry
VRP.setParam("Threads", 6) # Number of CPU threads used concurrently, my CPU has 12 threads
```

Figure 1. Advanced Gurobi Settings

Firstly, the code has been run for 2 hours and in the first 3 minutes it focused on finding an incumbent (upper bound) solution using Gurobi's NoRel heuristic. Moreover, solver has been designed to focus on gap improvement for the first 6300 minutes and then focuses on solution improvement for the rest of the time. Since the problem is an NP-hard problem and the mathematical formulation is a complex model; therefore, focusing on gap improvement was the logical approach. Additionally, tightening the bounds and adding aggressive cuts have been added to the code to increase the optimality gap reduction.

Table 1: Input parameters

Number of Nodes	21 (1 depot node and 10 shipment nodes)
Number of Days	2
Number of Vehicles	3 (small type, medium type, large type)
Vehicle Speed	60 km/h
Vehicle Capacity	22 (small), 30 (medium), 38 (large)
Fixed Vehicle Rental Cost	1750 AUD (small), 2000 AUD (medium), 2500 AUD (large)
Distance per km Cost	15 AUD (small), 18 AUD (medium), 22 AUD (large)
Cross-docking Cost	2500 AUD
Daily Distance Travel Limit	8 km
Reload/handling Time	30 minutes
Depot Capacity	200 pallets

Input parameters are provided on the above Table 1. Additionally, haversine distance formula was used as the calculation method to calculate the distance from one node to another. Furthermore, obtained distance matrix was divided to vehicle speed to calculate the travel time from one node to another.

Dataset Generation Approach:

Since no dataset has been provided by the industry partners, I have followed a synthetic dataset generation approach to test the established mathematical formulation. The generated dataset is an excel file and consists of two sheets. There is one depot which serves as a central point and works as a cross docking and inventory holding station. Furthermore, apart from the depot there are in total 20 shipment locations with 10 pickup and 10 delivery points. Moreover, 3 of the pickups of the shipments have been done before the horizon to include some pickups that are staying at the depot scenario in the dataset. Since the shipment horizon is two days, same logic has been implemented in the dataset. The depot works between 8 AM to 8 PM each day and time window values of the depot and the node locations have converted into minutes cumulatively. In addition, service time is selected as 15 minutes and it is fixed for every location in the dataset. Furthermore, the node quantities are selected wisely ranging from 8 pallets to 28 pallets. Lastly, screenshots of the generated dataset have been provided in the appendix section of the report. Also, developed code has been uploaded to my GitHub profile and it has been shared with the audience in the appendix section of the report for further usage and enhancements.

3.2 Theoretical / Numerical Results and Analysis

As indicated in the other sections of the report this problem is an NP-hard combinatorial optimisation problem. Various techniques have been implemented to obtain a feasible solution within an acceptable time frame. One of the most significant outcomes of this problem is the impact of performance optimisation techniques. Even though the code was set to run for 2 hours, solver found the optimal result in 859 seconds. This is a crucial indication that implemented techniques have worked effectively.

Established Routes with Mathematical Formulation — Objective Function Value: 26,089.89 AUD

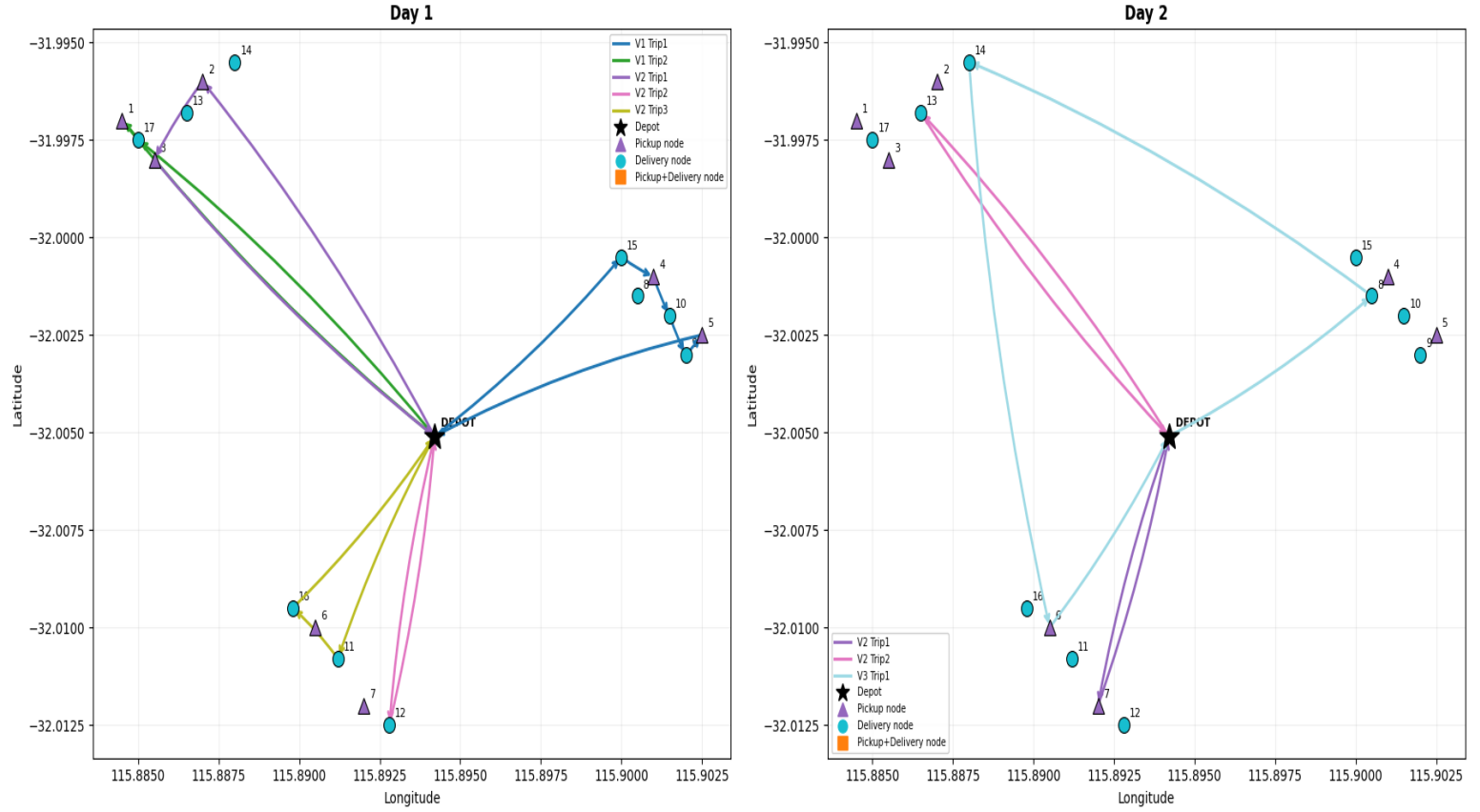


Figure 2. Visualization of the Established Routes

As shown in Figure 2 above, the established routes in the two days of shipment horizon are visualized to provide practical and user-friendly outcomes to the audience. All of the established routes, used vehicles and created trips are visualized by different colors. Moreover, node locations are shown on the map with different shapes to help the viewers. As presented in Figure 2, all of the available vehicles in the fleet are used to serve the customers within the shipment horizon. Vehicle 1 is only rented for day 1, vehicle 3 is also only rented for day 3, but vehicle 2 is used on both days to visit the node locations. Furthermore, total of 8 trip was needed to complete the routing operations. 2 trips were created by vehicle 1, vehicle 2 created 5 trips and vehicle 3 has only created 1 trip. In addition, as illustrated in Figure 2 above, the objective function of the mathematical model was found as 26,089.89 AUD. Daily fixed rental cost created an 8,250 AUD expense, distance per kilometer cost created a 339.89 AUD expense and cross docking cost constituted the majority of the total cost with contributing 17,500 AUD.

```

Optimal solution found (tolerance 1.00e-04)
Best objective 2.60898858376e+04, best bound 2.60898858376e+04, gap 0.0000%

Model status is: 2

Objective Function value is: 26089.88583755124
x[0,2,2,1,1] 1.000000
x[0,7,2,2,1] 1.000000
x[0,8,3,2,1] 1.000000
x[0,11,2,1,3] 1.000000
x[0,12,2,1,2] 1.000000
x[0,13,2,2,2] 1.000000
x[0,15,1,1,1] 1.000000
x[0,17,1,1,2] 1.000000
x[1,0,1,1,2] 1.000000
x[2,3,2,1,1] 1.000000
x[3,0,2,1,1] 1.000000
x[4,10,1,1,1] 1.000000
x[5,0,1,1,1] 1.000000
x[6,0,3,2,1] 1.000000
x[7,0,2,2,1] 1.000000
x[8,14,3,2,1] 1.000000
x[9,5,1,1,1] 1.000000
x[10,9,1,1,1] 1.000000
x[11,16,2,1,3] 1.000000
x[12,0,2,1,2] 1.000000
x[13,0,2,2,2] 1.000000
x[14,6,3,2,1] 1.000000
x[15,4,1,1,1] 1.000000
x[16,0,2,1,3] 1.000000
x[17,1,1,1,2] 1.000000
w[1,1,1] 1.000000
w[1,1,2] 1.000000
w[2,1,1] 1.000000
w[2,1,2] 1.000000
w[2,1,3] 1.000000
w[2,2,1] 1.000000
w[2,2,2] 1.000000
w[3,2,1] 1.000000
v[1,1] 1.000000
v[2,1] 1.000000
v[2,2] 1.000000
v[3,2] 1.000000

```

Figure 3. VS Code Terminal Output

```

b[0,1,1,1] 738.827250
b[0,1,1,2] 845.834337
b[0,2,1,1] 360.000000
b[0,2,1,2] 815.834337
b[0,2,1,3] 862.501064
b[0,2,2,1] 1560.000000
b[0,2,2,2] 1832.276448
b[0,3,2,1] 1635.794793
b[1,1,1,2] 1080.000000
b[2,2,1,1] 361.218547
b[3,2,1,1] 500.000000
b[4,1,1,1] 754.685547
b[5,1,1,1] 800.000000
b[6,3,2,1] 1816.629469
b[7,2,2,1] 1620.000000
b[8,3,2,1] 1636.511137
b[9,1,1,1] 784.927103
b[10,1,1,1] 769.806325
b[11,2,1,3] 863.195134
b[12,2,1,2] 816.667701
b[13,2,2,2] 1849.412665
b[14,3,2,1] 1800.000000
b[15,1,1,1] 739.576080
b[16,2,1,3] 878.390891
b[17,1,1,2] 847.045444
u[0,1,1,1] 14.000000
u[0,1,1,2] 14.000000
u[0,2,1,1] 9.000000
u[0,2,1,2] 18.000000
u[0,2,1,3] 26.000000
u[0,2,2,2] 22.000000
u[0,3,2,1] 38.000000
u[1,1,1,2] 10.000000
u[2,2,1,1] 21.000000
u[3,2,1,1] 30.000000
u[4,1,1,1] 21.000000
u[5,1,1,1] 18.000000
u[6,3,2,1] 22.000000
u[7,2,2,1] 28.000000
u[8,3,2,1] 28.000000
u[10,1,1,1] 12.000000
u[11,2,1,3] 11.000000
u[15,1,1,1] 6.000000

```

Figure 4. VS Code Terminal Output

```

l_out[1,1,1] 14.000000
l_out[1,1,2] 14.000000
l_out[2,1,1] 9.000000
l_out[2,1,2] 18.000000
l_out[2,1,3] 26.000000
l_out[2,2,2] 22.000000
l_out[3,2,1] 38.000000
l_in[1,1,1] 18.000000
l_in[1,1,2] 10.000000
l_in[2,1,1] 30.000000
l_in[2,2,1] 28.000000
l_in[3,2,1] 22.000000
delta[0] 1.000000
delta[1] 1.000000
delta[2] 1.000000
delta[3] 1.000000
delta[4] 1.000000
delta[5] 1.000000
delta[6] 1.000000
I[1] 10.000000

Established Routes:
Vehicle 1 | Day 1 | Trip 1 | Route: [0, 15, 4, 10, 9, 5, 0]
Vehicle 1 | Day 1 | Trip 2 | Route: [0, 17, 1, 0]
Vehicle 2 | Day 1 | Trip 1 | Route: [0, 2, 3, 0]
Vehicle 2 | Day 1 | Trip 2 | Route: [0, 12, 0]
Vehicle 2 | Day 1 | Trip 3 | Route: [0, 11, 16, 0]
Vehicle 2 | Day 2 | Trip 1 | Route: [0, 7, 0]
Vehicle 2 | Day 2 | Trip 2 | Route: [0, 13, 0]
Vehicle 3 | Day 2 | Trip 1 | Route: [0, 8, 14, 6, 0]

Route Summary:
Vehicle 1 | Day 1 | Trip 1 | Distance: 2.0071 km | Load Out: 14.00 | Cap: 22
Vehicle 1 | Day 1 | Trip 2 | Distance: 2.5677 km | Load Out: 14.00 | Cap: 22
Vehicle 2 | Day 1 | Trip 1 | Distance: 2.6207 km | Load Out: 9.00 | Cap: 30
Vehicle 2 | Day 1 | Trip 2 | Distance: 1.6667 km | Load Out: 18.00 | Cap: 30
Vehicle 2 | Day 1 | Trip 3 | Distance: 1.5313 km | Load Out: 26.00 | Cap: 30
Vehicle 2 | Day 2 | Trip 1 | Distance: 1.5896 km | Load Out: 0.00 | Cap: 30
Vehicle 2 | Day 2 | Trip 2 | Distance: 2.3486 km | Load Out: 22.00 | Cap: 30
Vehicle 3 | Day 2 | Trip 1 | Distance: 4.3473 km | Load Out: 38.00 | Cap: 38

```

Figure 5. VS Code Terminal Output

```

Shipment Outcomes:
Shipment 1 [p=1, d=8, q=10]: CROSS-DOCKING (different vehicle, different day)
Pickup : V1 | Day 1 | Trip 2 | b=1080.0 TW=[1080, 1140] OK
Delivery: V3 | Day 2 | Trip 1 | b=1636.5 TW=[1560, 1680] OK
Predecessor: pickup Day 1 <= delivery Day 2 - OK
Shipment 2 [p=2, d=9, q=12]: CROSS-DOCKING (different vehicle, same day)
Pickup : V2 | Day 1 | Trip 1 | b=361.2 TW=[360, 500] OK
Delivery: V1 | Day 1 | Trip 1 | b=784.9 TW=[700, 1200] OK
Predecessor: pickup Day 1 <= delivery Day 1 - OK
Shipment 3 [p=3, d=10, q=9]: CROSS-DOCKING (different vehicle, same day)
Pickup : V2 | Day 1 | Trip 1 | b=500.0 TW=[360, 500] OK
Delivery: V1 | Day 1 | Trip 1 | b=769.8 TW=[700, 1200] OK
Predecessor: pickup Day 1 <= delivery Day 1 - OK
Shipment 4 [p=4, d=11, q=15]: CROSS-DOCKING (different vehicle, same day)
Pickup : V1 | Day 1 | Trip 1 | b=754.7 TW=[360, 800] OK
Delivery: V2 | Day 1 | Trip 3 | b=863.2 TW=[420, 950] OK
Predecessor: pickup Day 1 <= delivery Day 1 - OK
Shipment 5 [p=5, d=12, q=18]: CROSS-DOCKING (different vehicle, same day)
Pickup : V1 | Day 1 | Trip 1 | b=800.0 TW=[360, 800] OK
Delivery: V2 | Day 1 | Trip 2 | b=816.7 TW=[420, 950] OK
Predecessor: pickup Day 1 <= delivery Day 1 - OK
Shipment 6 [p=6, d=13, q=22]: CROSS-DOCKING (different vehicle, same day)
Pickup : V3 | Day 2 | Trip 1 | b=1816.6 TW=[1560, 1900] OK
Delivery: V2 | Day 2 | Trip 2 | b=1849.4 TW=[1750, 2100] OK
Predecessor: pickup Day 2 <= delivery Day 2 - OK
Shipment 7 [p=7, d=14, q=28]: CROSS-DOCKING (different vehicle, same day)
Pickup : V2 | Day 2 | Trip 1 | b=1620.0 TW=[1620, 1950] OK
Delivery: V3 | Day 2 | Trip 1 | b=1800.0 TW=[1800, 2200] OK
Predecessor: pickup Day 2 <= delivery Day 2 - OK
Shipment 8 [p=depot, d=15, q=8]: Direct (pre-positioned at depot)
Delivery: V1 | Day 1 | Trip 1 | b=739.6 TW=[420, 950] OK
Shipment 9 [p=depot, d=16, q=11]: Direct (pre-positioned at depot)
Delivery: V2 | Day 1 | Trip 3 | b=878.4 TW=[420, 950] OK
Shipment 10 [p=depot, d=17, q=14]: Direct (pre-positioned at depot)
Delivery: V1 | Day 1 | Trip 2 | b=847.0 TW=[420, 950] OK

Depot Inventory Levels:
Day 0 (initial I0): 33.00
End of Day 1: 10.00

End of Day 2: 0.00

```

Figure 6. VS Code Terminal Output

VS Code Terminal Outcomes have been provided in Figures 3, 4, 5 and 6. It shows the outcome of the mathematical model with providing the results of the objective function value and the decision variables. Moreover, extra printing sections such as established routes, route summary and shipment outcomes have been presented to the viewers to facilitate their analyzing process. Loads of vehicles, nodes and the depot can be monitored easily. As shown in the figures, the shipment process started with 33 pallets which were the shipments that have already been picked up before the horizon. After the first day routing, depot's inventory level was 10 pallets and it transferred to day 2. In the second day, all of the routing operations have been performed without leaving any pallets as an inventory to the depot. Furthermore, node visiting times are provided with the users to monitor the timely deliveries. Lastly, cross docking scenarios are very significant for the solution of this mathematical model. The optimal routing operation has provided 7 cross docking operations and 3 direct shipments during the horizon. The shipments that are already waiting at the depot have been delivered directly without any cross docking. On the other hand, 6 shipments have been performed with cross docking operations on the same day. Additionally, there is a one cross docking scenario that has happened between days. Pickup of shipment 1 is made on day one and its delivery is made on day two by a different vehicle. Consequently, this mathematical model has provided a solution methodology for a highly specific vehicle routing problem but further developments must be needed to make this study in industry standards and research level.

4. Conclusion and Future Work

4.1 Significance of Findings

This project has presented a mathematical model to minimize the total cost of freight operations. Developed mathematical formulation has included various VRP extensions such as heterogenous fleet, capacity limitations, time window requirements, vehicle renting selection, daily distance limitation, multi-trip option, cross docking scenarios, inventory management and pickup and delivery synchronization. In addition, visualization of the established routes has been provided to users to facilitate the monitoring. Moreover, it has been shown that implementing advanced performance optimisation techniques such as tight big-M selection for each constraint that has big-M, arc pre-elimination method before running the mathematical model code and selecting Gurobi settings wisely improve the computation time and enhances the optimality gap reduction crucially. Even though the problem type is NP-hard and mathematical formulation is highly complex, this project has demonstrated that commercial

optimisation solver tool Gurobi provides optimal solution within acceptable time frame for small dataset. Most significantly, it has been confirmed that established mathematical model provides efficient routing strategies and can be used by industry partners to optimize their freight operations.

4.2 Recommendations and Future Work

As examined in the methodology and results section of the report the main disadvantage of the problem is its complexity. VRP is a well-known NP-hard problem and this problem is a highly complex extension of a VRP; therefore, providing optimal routing strategies for real life large datasets is a significant topic to focus on. Commercial optimisation solver tools such as Gurobi and CPLEX are most popular solvers for mathematical formulations. Furthermore, underlying algorithms in those commercial solver tools are developing rapidly and providing optimal results for non-complex mathematical models, however their efficiency is still limited for NP-hard problems and huge datasets. Various metaheuristics such as Particle Swarm Optimisation, Adaptive Large Neighborhood Search, Simulated Annealing and Genetic Algorithm can be used as the primary solution algorithm for large instances. Following my discussions with my project supervisor, I am planning to continue the project next semester and I want to expand the scope of the project. Although I am graduating this semester, I will continue the project under Elham's guidance and I will consider this process as an opportunity to improve my knowledge and skillset in operations research. First thing I will focus on next semester is to decompose the problem into smaller parts to enhance the computation speed. The planning horizon is 2 days, hence decomposing the problem into two sub models as day one and day two and solving each problem separately will reduce the computation speed. After that, obtained outcomes can be given to the Gurobi code of the main mathematical formulation as a warm start and it will accelerate the optimality gap reduction significantly. Secondly, I will develop a genetic algorithm to solve the problem for large datasets. Rather than using the developed genetic algorithm libraries in Python, I will design and code my genetic algorithm from scratch because it is a unique version of a VRP. Since my problem is a complex extension of a VRP and it includes various problem requirements, encoding the chromosome structure will be crucial to develop an effective genetic algorithm. Furthermore, initial population generation, crossover and mutation operations are mandatory functions of the algorithm to obtain good quality solutions. Advanced genetic algorithm techniques such as adaptive mutation rate, elitism, perturbation and diversity-based stopping will be implemented to the algorithm to expand the scope of the search space and avoid tackling local optimums. In addition, testing the

mathematical model code with a computer which has better RAM and CPU configurations can reduce the computation time. Furthermore, rather than using synthetic data, a real dataset can be gathered and the mathematical model code and the genetic algorithm to be developed can be tested with a real dataset. Moreover, rather than using a python as the programming language, a compiled language such as C++ can be used to code the mathematical model and genetic algorithm to increase the computation speed. The developed codes will be available in my GitHub profile and it can be shared with computer scientists to make data structures and algorithms-based revisions. Lastly, the developed mathematical model is a complex extension of a VRP and it covers various concepts; therefore, it can be used as a template to provide solution methodologies for similar vehicle routing problems by different audiences. Even though the project has focused on optimizing routing strategies for road transportation, it can easily be converted into air and maritime transportation problems and this mathematical model can be used as a skeleton for developing air and maritime mathematical formulations.

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6. Appendix

GitHub Link: <https://github.com/CemAlpaslan23/Routing-Optimisation-for-Freight-Operations-Project>

	A	B	C	D	E	F	G
1	NodeID	Longitude	Latitude	Earliest	Latest	ServiceTime	Type
2	0	115,8942	-32,0051	360	1200	0	depot
3	1	115,8845	-31,997	1080	1140	15	pickup
4	2	115,887	-31,996	360	500	15	pickup
5	3	115,8855	-31,998	360	500	15	pickup
6	4	115,901	-32,001	360	800	15	pickup
7	5	115,9025	-32,0025	360	800	15	pickup
8	6	115,8905	-32,01	1560	1900	15	pickup
9	7	115,892	-32,012	1620	1950	15	pickup
10	8	115,9005	-32,0015	1560	1680	15	delivery
11	9	115,902	-32,003	700	1200	15	delivery
12	10	115,9015	-32,002	700	1200	15	delivery
13	11	115,8912	-32,0108	420	950	15	delivery
14	12	115,8928	-32,0125	420	950	15	delivery
15	13	115,8865	-31,9968	1750	2100	15	delivery
16	14	115,888	-31,9955	1800	2200	15	delivery
17	15	115,9	-32,0005	420	950	15	delivery
18	16	115,8898	-32,0095	420	950	15	delivery
19	17	115,885	-31,9975	420	950	15	delivery

	A	B	C	D	E
1	ShipmentID	PickupNode	DeliveryNode	Quantity	AlreadyPickedUp
2	1	1	8	10	0
3	2	2	9	12	0
4	3	3	10	9	0
5	4	4	11	15	0
6	5	5	12	18	0
7	6	6	13	22	0
8	7	7	14	28	0
9	8	0	15	8	1
10	9	0	16	11	1
11	10	0	17	14	1

Figure 7. Generated Dataset